

Data Assimilation in Nonlinear Fluid Flow II

The Ensemble Kalman Filter: nonlinear forecasts, sampled covariance, and computational cost

Presentation 2 of 4

1. Learning objectives

- Replace the dense KF covariance by an ensemble covariance.
- Derive the forecast and analysis steps of the stochastic EnKF.
- Understand sampling error, rank deficiency, inflation, and localization.
- Show that nonlinear Burgers assimilation becomes accurate—but requires many model forecasts.

2. Recall the problem from presentation 1

The plain KF needs

$$\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{w}_k.$$

But our fluid model is nonlinear,

$$\mathbf{u}^{k+1} = \mathcal{M}_{\Delta t}(\mathbf{u}^k).$$

The frozen linear KF achieved mean RMSE **0.647**.

Key idea

We want to use the actual nonlinear forecast model without carrying a dense $n \times n$ covariance.

3. Core idea of the EnKF

Represent uncertainty by M state samples,

$$\{\mathbf{x}^{(i)}\}_{i=1}^M.$$

Estimate mean and covariance from the ensemble,

$$\bar{\mathbf{x}} = \frac{1}{M} \sum_{i=1}^M \mathbf{x}^{(i)},$$
$$\mathbf{P}_e = \frac{1}{M-1} \sum_{i=1}^M (\mathbf{x}^{(i)} - \bar{\mathbf{x}})(\mathbf{x}^{(i)} - \bar{\mathbf{x}})^\top.$$

No tangent model and no explicit covariance propagation are required.

4. Nonlinear ensemble forecast

Each member is propagated independently,

$$\mathbf{x}_k^{f,(i)} = \mathcal{M}_{\Delta t}(\mathbf{x}_{k-1}^{a,(i)}) + \mathbf{w}_k^{(i)}.$$

Then

$$\bar{\mathbf{x}}_k^f = \frac{1}{M} \sum_i \mathbf{x}_k^{f,(i)}, \quad \mathbf{P}_k^f \approx \frac{1}{M-1} \mathbf{X}_k' \mathbf{X}_k'^{\top}.$$

The nonlinear model shapes the forecast cloud directly.

5. Ensemble anomalies

Define the anomaly matrix

$$\mathbf{X}' = \begin{bmatrix} \mathbf{x}^{(1)} - \bar{\mathbf{x}} & \dots & \mathbf{x}^{(M)} - \bar{\mathbf{x}} \end{bmatrix} \in \mathbb{R}^{n \times M}.$$

Then

$$\mathbf{P}_e = \frac{1}{M-1} \mathbf{X}' \mathbf{X}'^\top.$$

This factorized form is important: algorithms can often work with $\mathbf{X}'$ without explicitly forming a dense covariance.

6. Stochastic EnKF analysis

With linear observations $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{v}$,

$$\mathbf{K}_e = \mathbf{P}_e^f \mathbf{H}^\top (\mathbf{H} \mathbf{P}_e^f \mathbf{H}^\top + \mathbf{R})^{-1}.$$

Perturb the observation for each member,

$$\mathbf{y}^{(i)} = \mathbf{y} + \mathbf{v}^{(i)}, \quad \mathbf{v}^{(i)} \sim \mathcal{N}(0, \mathbf{R}),$$

and update

$$\mathbf{x}^{a,(i)} = \mathbf{x}^{f,(i)} + \mathbf{K}_e (\mathbf{y}^{(i)} - \mathbf{H}\mathbf{x}^{f,(i)}).$$

7. Why perturb the observations?

If every member used the identical measurement $\mathbf{y}$, the analysis ensemble would be too narrow in the stochastic EnKF formulation.

Purpose

Observation perturbations make the ensemble analysis covariance consistent with the Kalman covariance in the large-ensemble linear–Gaussian limit.

Deterministic square-root EnKFs avoid random perturbations by transforming anomalies directly.

8. Deterministic alternatives

Common families include:

- Ensemble Transform Kalman Filter (ETKF),
- Ensemble Square-Root Filter (EnSRF),
- Local Ensemble Transform Kalman Filter (LETKF).

They seek an analysis ensemble with the desired mean/covariance without adding sampled observation noise.

Key idea

This presentation uses stochastic EnKF because the algebra and code are especially transparent.

9. Minimal EnKF code pattern

```
X = nonlinear_forecast(X, dt, nu)
A = X - X.mean(axis=0)
P = A.T @ A / (M - 1)
K = solve(H @ P @ H.T + R, H @ P).T
Y = y[k] + normal(0, obs_std, size=(M, m))
X = X + (Y - X[:, obs_idx]) @ K.T
```

The full implementation is in `code/da_methods.py`.

10. Theorem: EnKF recovers KF statistics as $M \rightarrow \infty$

Idealized linear result

For a linear–Gaussian system, independent ensemble members with finite second moments satisfy

$$\bar{\mathbf{x}}_M^f \rightarrow \mathbb{E}[\mathbf{x}^f], \quad \mathbf{P}_{e,M}^f \rightarrow \text{Cov}(\mathbf{x}^f)$$

almost surely as $M \rightarrow \infty$; the perturbed-observation EnKF analysis then converges to the KF analysis statistics.

The statement is about an idealized large-ensemble limit; finite- M filters exhibit sampling error.

11. Proof sketch: law of large numbers

For independent forecast samples,

$$\frac{1}{M} \sum_{i=1}^M \mathbf{x}^{(i)} \xrightarrow{a.s.} \mathbb{E}[\mathbf{x}],$$

and each covariance entry is an empirical average of finite-moment random variables, so

$$\mathbf{P}_{e,M} \xrightarrow{a.s.} \mathbf{P}.$$

The Kalman gain is a continuous function of $\mathbf{P}$ whenever the innovation covariance is nonsingular. Therefore

$$\mathbf{K}_{e,M} \rightarrow \mathbf{K}.$$

Applying the same argument after the analysis update completes the sketch. $\square$

12. Sampling error scales slowly

For fixed state dimension and finite fourth moments, Monte Carlo estimates typically obey

$$\bar{\mathbf{x}}_M - \mathbb{E}[\mathbf{x}] = \mathcal{O}_p(M^{-1/2}),$$

and individual covariance entries have the same root- M sampling rate.

Consequence

Reducing sampling noise by a factor of 2 can require roughly 4 times as many ensemble members.

13. Rank limitation

Because the anomaly columns sum to zero,

$$\text{rank}(\mathbf{X}') \leq M - 1,$$

therefore

$$\text{rank}(\mathbf{P}_e) \leq M - 1.$$

If $n \gg M$, the EnKF covariance lives in a low-dimensional subspace.

Important

In geophysical and CFD problems, n may be millions while M may only be tens or hundreds.

14. Spurious long-range correlations

A finite ensemble can create accidental correlations between distant state variables. A common remedy is covariance localization,

$$\mathbf{P}_{\text{loc}} = \mathbf{C} \circ \mathbf{P}_e,$$

where $\circ$ is the Hadamard product and $\mathbf{C}$ decays with spatial distance.

Key idea

Localization injects the prior belief that very distant points should not be strongly correlated unless physics supports it.

15. Inflation prevents ensemble collapse

A common multiplicative inflation is

$$\mathbf{x}^{(i)} \leftarrow \bar{\mathbf{x}} + \lambda(\mathbf{x}^{(i)} - \bar{\mathbf{x}}), \quad \lambda > 1.$$

It compensates for under-dispersive ensembles caused by:

- finite- M sampling,
- unrepresented model error,
- repeated assimilation updates.

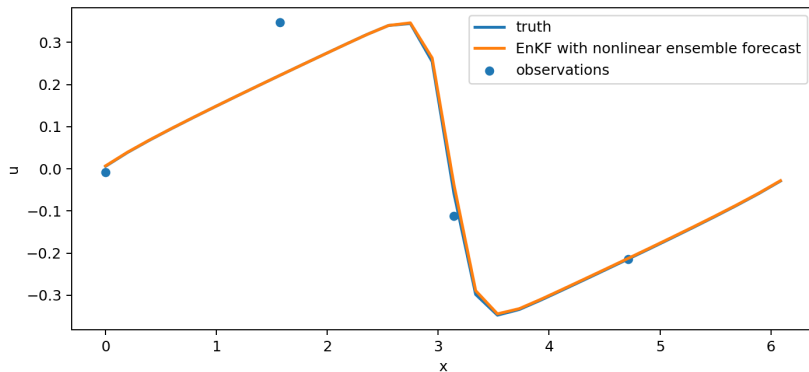
Our tutorial uses $\lambda = 1.02$.

16. Same nonlinear experiment as presentation 1

Truth model	nonlinear viscous Burgers equation
Grid	$N = 32$
Observations	every 8th point
Noise	$\sigma_y = 0.08$
Ensemble	$M = 40$
Inflation	1.02

The important change is that **every ensemble member is forecast with the nonlinear model.**

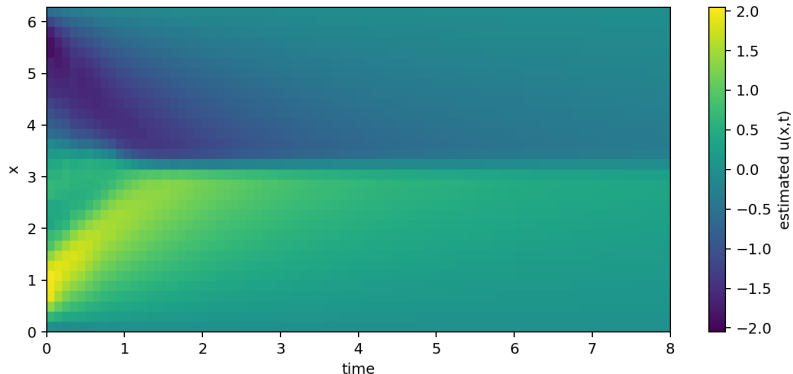
17. EnKF final-state reconstruction



Key idea

With nonlinear member forecasts, the analysis follows the true final flow profile closely even though only four grid locations are observed.

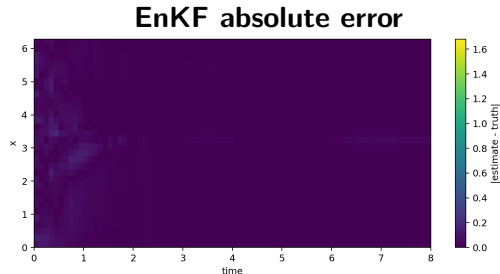
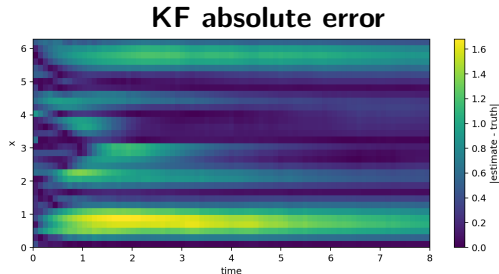
18. EnKF output over the complete trajectory



Key idea

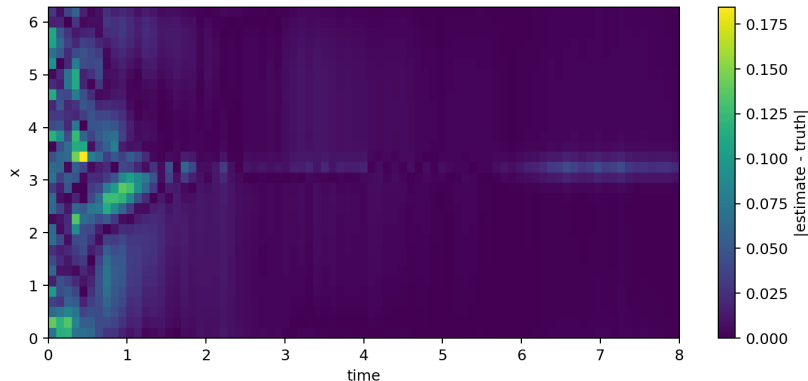
The reconstructed space-time field follows the moving nonlinear structures rather than only matching the sensors at analysis times.

19. Error maps: KF versus EnKF on the same color scale



Both figures use the same error range; each image includes its colorbar. The large coherent KF error almost disappears when the nonlinear dynamics are propagated by the ensemble.

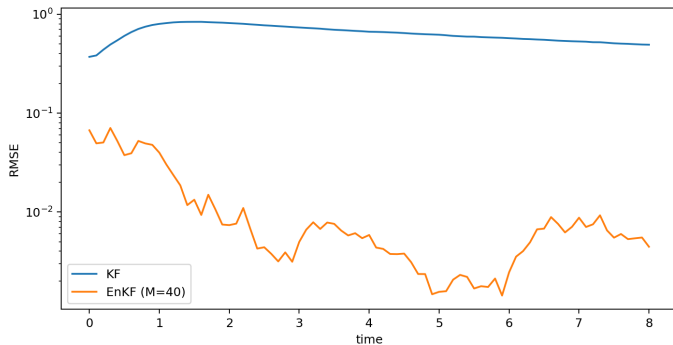
20. EnKF residual error—zoomed color scale



Key idea

A method-specific colorbar reveals the much smaller residual structure that is hidden on the common KF/EnKF scale.

21. RMSE: linear KF versus nonlinear EnKF



Method	Mean RMSE	Forecast physics
KF	0.647	frozen linear model
EnKF, $M = 40$	0.012	nonlinear Burgers model

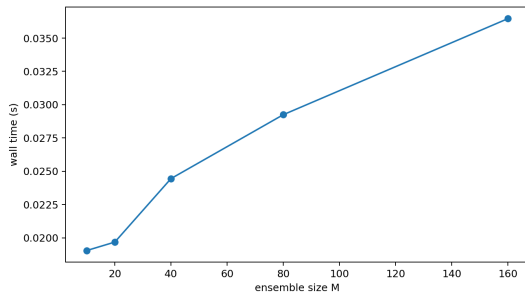
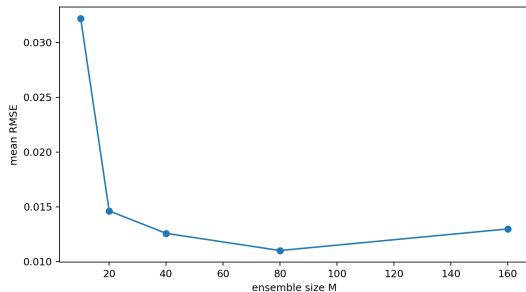
22. Why the EnKF succeeds in this experiment

- ① Every member undergoes nonlinear advection and diffusion.
- ② The ensemble cloud develops state-dependent, flow-dependent error directions.
- ③ Sample cross-covariances connect the four sensors to unobserved grid points.
- ④ Repeated observations recenter the ensemble around the measured trajectory.

Key idea

The improvement comes from using the nonlinear forecast geometry while estimating covariance from samples.

23. Ensemble size: accuracy versus cost



Important

Monte Carlo error decreases only gradually with M , while nonlinear forecast work grows approximately linearly with M .

24. Structural EnKF cost

Let $C_{\mathcal{M}}(n)$ be the cost of one nonlinear forecast. A rough cycle costs

$$C_{\text{EnKF}} \approx M C_{\mathcal{M}}(n) + C_{\text{statistics}}(n, M, m).$$

For $M = 40$,

40 nonlinear trajectories are advanced every assimilation cycle.

- Parallelism can reduce wall time.
- Parallelism does not remove total compute, memory, or communication.
- Large M improves sampling but directly increases forecast cost.

25. KF versus EnKF: the tradeoff made explicit

Property	KF	EnKF
Forecast dynamics	fixed linear F	nonlinear $\mathcal{M}$ per member
Covariance	dense deterministic P	sampled low-rank P_e
Storage	$\mathcal{O}(n^2)$	$\mathcal{O}(nM)$
Forecast solves/cycle	1 linear solve	M nonlinear solves
Sampling error	none	yes, $\sim M^{-1/2}$
High-dimensional scaling	covariance bottleneck	ensemble-cost bottleneck
Tutorial RMSE	0.647	0.012

26. When is the EnKF worth the computational cost?

EnKF is attractive when	EnKF is difficult when
Nonlinear flow dependence matters strongly	Each nonlinear forecast is extremely expensive
A posterior spread / uncertainty estimate is valuable	Memory cannot hold tens-hundreds of states
Forecasts can run efficiently in parallel	Communication dominates parallel execution
Localization structure is physically meaningful	Small M creates severe spurious correlations
Accuracy justifies repeated model evaluation	Real-time latency has a strict budget

Key idea

The central tradeoff is better nonlinear, flow-dependent estimation in exchange for repeated model solves and finite-ensemble sampling error.

27. Main takeaways

- The EnKF propagates uncertainty through nonlinear dynamics by evolving an ensemble of model states.
- Sample mean and covariance replace the dense covariance recursion of the classical KF.
- In the Burgers example, mean RMSE decreases from 0.647 to **0.012**.
- The common-scale error maps make the improvement spatially obvious, not only visible in a scalar metric.
- The price is M nonlinear forecasts per cycle, low-rank covariance, sampling noise, and tuning through localization/inflation.